

Project 1: Exploratory Data Analysis and Simple Regression

CSCI 4360 – Data Science ML II

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1 Introduction

This project examines three datasets using exploratory data analysis and simple linear regression: AutoMPG, Concrete Compressive Strength, and Airfoil Self-Noise. For each dataset, the analysis checks string-valued variables, missing values, and potential outliers; summarizes the variables; examines Pearson correlations; selects the two predictors with the strongest absolute correlations with the response; and fits separate simple linear regression models in both Python `statsmodels` and ScalaTion.

Potential outliers were identified using the $1.5 \times IQR$ rule. An observation flagged by this rule was not automatically removed; the decision to retain or remove it was based on whether there was evidence that the value was invalid. The regression model used throughout the project is

$$\hat{y} = b_0 + b_1x,$$

where b_0 is the intercept and b_1 is the estimated change in the response associated with a one-unit increase in the predictor.

2 AutoMPG

The AutoMPG dataset originally contained 398 observations and nine variables, including the response variable, miles per gallon (MPG). The goal was to identify the two numerical predictors with the strongest linear associations with MPG and compare simple regression results across `statsmodels` and `ScalaTion`.

2.1 Preprocessing

The original data included the string-valued variable `car_name` and the categorical variable `origin`, whose categories were represented by the integers 1–3. These variables were excluded from the numerical correlation and simple regression analysis because they were not continuous numerical predictors.

Six observations had missing values in `horsepower`, corresponding to approximately 1.51% of the original dataset. These six rows were removed, leaving 392 observations for the numerical analysis.

Potential outliers were examined with boxplots and the $1.5 \times IQR$ rule. Ten potential outliers were identified for horsepower and eleven for acceleration. No potential outliers were identified for MPG, cylinders, displacement, weight, or model year. The flagged observations were retained because the values were plausible and there was no evidence of data-entry errors.

2.2 Statistical Summary

Table 1 summarizes MPG and all retained numerical predictors after preprocessing.

Table 1: AutoMPG descriptive statistics after preprocessing.

Variable	Count	Mean	SD	Min	Q1	Median	Q3	Max
MPG	392	23.446	7.805	9.0	17.000	22.750	29.000	46.6
Cylinders	392	5.472	1.706	3.0	4.000	4.000	8.000	8.0
Displacement	392	194.412	104.644	68.0	105.000	151.000	275.750	455.0
Horsepower	392	104.469	38.491	46.0	75.000	93.500	126.000	230.0
Weight	392	2977.584	849.403	1613.0	2225.250	2803.500	3614.750	5140.0
Acceleration	392	15.541	2.759	8.0	13.775	15.500	17.025	24.8
Model Year	392	75.980	3.684	70.0	73.000	76.000	79.000	82.0

The retained variables show substantial variability, particularly in displacement, horsepower, and weight. MPG also varies considerably across the sample. Displacement, horsepower, and weight have means above their medians, which is consistent with some right-skewness in these variables.

2.3 Correlation Analysis and Predictor Selection

Table 2.2 gives the Pearson correlation matrix for the processed numeric dataset. Weight has the strongest absolute correlation with MPG ($r = -0.832$), followed by displacement ($r = -0.805$). These two variables were therefore selected for simple regression.

Table 2: AutoMPG Pearson correlation matrix.

	MPG	Cyl	Disp	HP	Wt	Acc	Year
MPG	1.000	-0.778	-0.805	-0.778	-0.832	0.423	0.581
Cyl	-0.778	1.000	0.951	0.843	0.898	-0.505	-0.346
Disp	-0.805	0.951	1.000	0.897	0.933	-0.544	-0.370
HP	-0.778	0.843	0.897	1.000	0.865	-0.689	-0.416
Wt	-0.832	0.898	0.933	0.865	1.000	-0.417	-0.309
Acc	0.423	-0.505	-0.544	-0.689	-0.417	1.000	0.290
Year	0.581	-0.346	-0.370	-0.416	-0.309	0.290	1.000

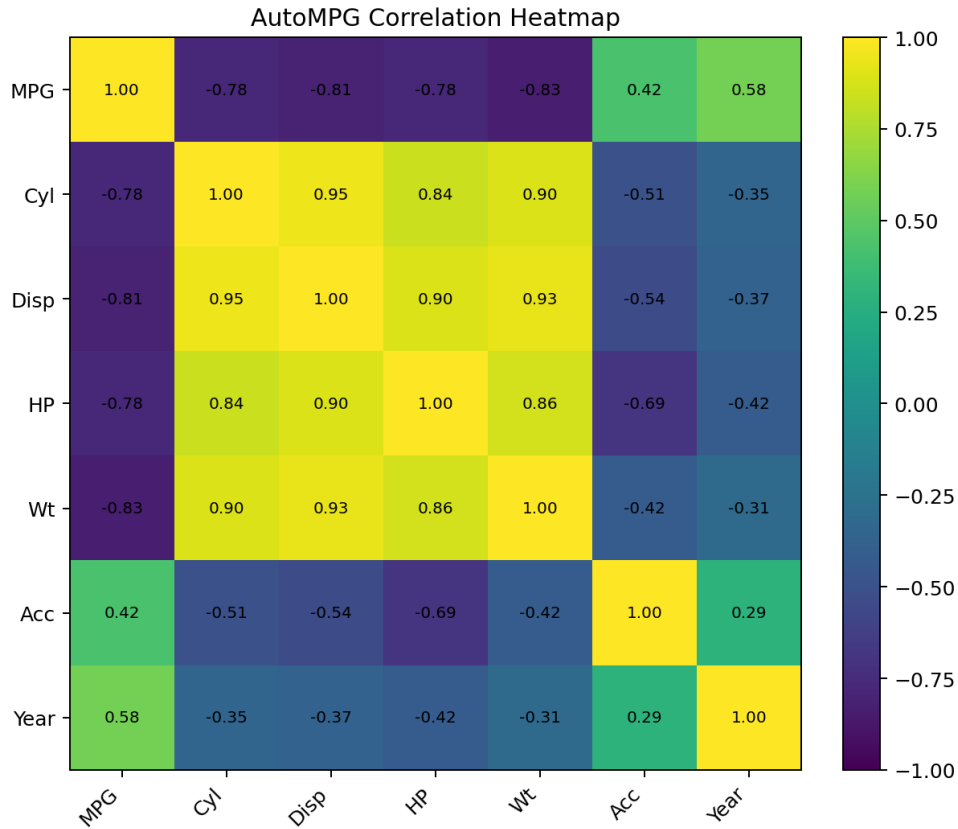


Figure 1: AutoMPG correlation heatmap. The strongest absolute correlations with MPG occur for weight and displacement.

2.4 Simple Regression Using Displacement

Simple linear regression was performed in both Python using `statsmodels` and Scala using `Scala-Tion`. The two implementations produced essentially identical results. The fitted equation was

$$\widehat{MPG} = 35.1206 - 0.0601(\text{Displacement}).$$

The negative slope indicates that greater engine displacement is associated with lower fuel economy. A one-unit increase in displacement is associated with an estimated decrease of approximately 0.0601 MPG. The slope was statistically significant ($p < 0.001$).

ScalaTion reported $R^2 = 0.648229$ and adjusted $R^2 = 0.647327$, consistent with the `statsmodels` results. Thus, displacement alone explained approximately 64.8% of the observed variation in MPG. ScalaTion also reported an RMSE of approximately 4.62 MPG.

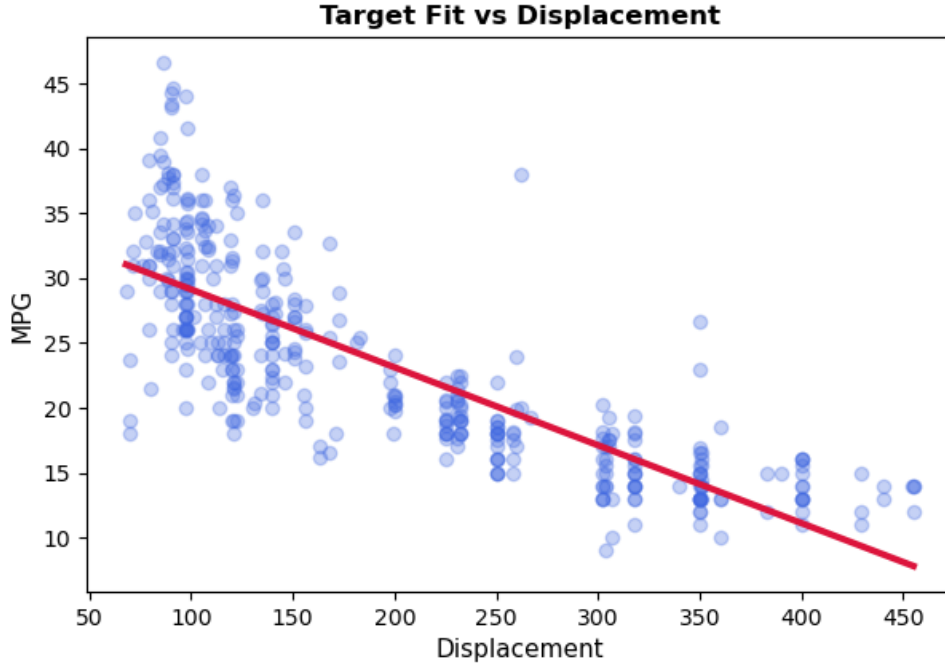


Figure 2: Observed MPG values and fitted simple regression values versus displacement.

2.5 Simple Regression Using Weight

The simple regression using vehicle weight also produced nearly identical results in ScalaTion and `statsmodels`. The fitted equation was

$$\widehat{MPG} = 46.2165 - 0.007647(\text{Weight}).$$

The negative slope indicates that heavier vehicles tend to have lower fuel economy. A one-unit increase in vehicle weight is associated with an estimated decrease of approximately 0.00765 MPG. The slope was statistically significant ($p < 0.001$).

ScalaTion reported $R^2 = 0.692630$ and adjusted $R^2 = 0.691842$, again matching the `statsmodels` results. Therefore, vehicle weight alone explained approximately 69.3% of the variation in MPG. ScalaTion reported an RMSE of approximately 4.32 MPG.

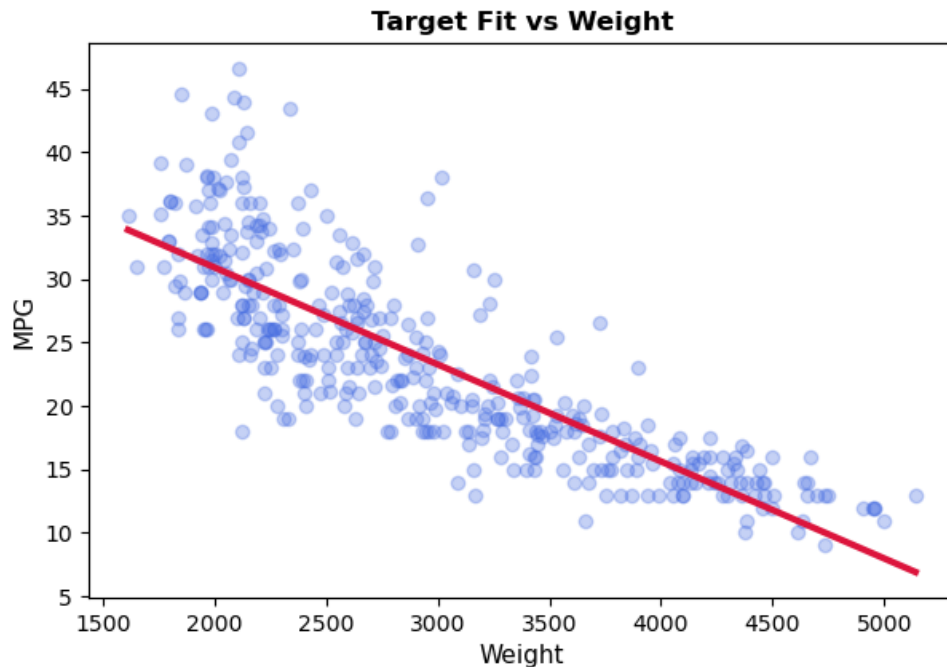


Figure 3: Observed MPG values and fitted simple regression values versus vehicle weight.

2.6 AutoMPG Discussion

Both selected predictors exhibited strong and statistically significant negative relationships with MPG, and the `ScalaTion` and `statsmodels` implementations produced essentially identical parameter estimates and goodness-of-fit measures. Vehicle weight provided the stronger of the two simple regression models, with $R^2 \approx 0.693$, compared with $R^2 \approx 0.648$ for displacement. This is consistent with the correlation analysis, in which weight also had the stronger absolute correlation with MPG.

3 Concrete Compressive Strength

The Concrete Compressive Strength dataset contains 1,030 observations, eight predictors, and one response variable. The response is concrete compressive strength, measured in megapascals (MPa).

3.1 Preprocessing

All nine variables were numeric, so no string-valued columns required removal or encoding. No missing values were identified, and therefore no imputation or row removal was required.

Potential outliers were evaluated with boxplots and the $1.5 \times IQR$ rule. The analysis identified 2 potential outliers for blast furnace slag, 9 for water, 10 for superplasticizer, 5 for fine aggregate, 59 for age, and 4 for compressive strength. Cement, fly ash, and coarse aggregate had no IQR outliers. The flagged observations were retained because the IQR criterion identifies unusual values but does not establish that those values are incorrect. In particular, large age values may correspond to valid longer curing periods.

3.2 Statistical Summary

Table 3: Concrete descriptive statistics.

Variable	Count	Mean	SD	Min	Q1	Median	Q3	Max
Cement	1030	281.168	104.506	102.00	192.375	272.900	350.000	540.0
Blast Furnace Slag	1030	73.896	86.279	0.00	0.000	22.000	142.950	359.4
Fly Ash	1030	54.188	63.997	0.00	0.000	0.000	118.300	200.1
Water	1030	181.567	21.354	121.80	164.900	185.000	192.000	247.0
Superplasticizer	1030	6.205	5.974	0.00	0.000	6.400	10.200	32.2
Coarse Aggregate	1030	972.919	77.754	801.00	932.000	968.000	1029.400	1145.0
Fine Aggregate	1030	773.580	80.176	594.00	730.950	779.500	824.000	992.6
Age	1030	45.662	63.170	1.00	7.000	28.000	56.000	365.0
Compressive Strength	1030	35.818	16.706	2.33	23.710	34.445	46.135	82.6

Several variables show evidence of skewness, particularly blast furnace slag, fly ash, superplasticizer, and age, as reflected by differences between their means and medians.

3.3 Correlation Analysis and Predictor Selection

Cement had the strongest absolute correlation with compressive strength ($r = 0.498$), followed by superplasticizer ($r = 0.366$). These variables were selected for the two simple regression analyses.

Table 4: Concrete Pearson correlation matrix.

	Cem	Slag	Ash	Water	SP	Coarse	Fine	Age	Strength
Cem	1.000	-0.275	-0.397	-0.082	0.092	-0.109	-0.223	0.082	0.498
Slag	-0.275	1.000	-0.324	0.107	0.043	-0.284	-0.282	-0.044	0.135
Ash	-0.397	-0.324	1.000	-0.257	0.378	-0.010	0.079	-0.154	-0.106
Water	-0.082	0.107	-0.257	1.000	-0.658	-0.182	-0.451	0.278	-0.290
SP	0.092	0.043	0.378	-0.658	1.000	-0.266	0.223	-0.193	0.366
Coarse	-0.109	-0.284	-0.010	-0.182	-0.266	1.000	-0.178	-0.003	-0.165
Fine	-0.223	-0.282	0.079	-0.451	0.223	-0.178	1.000	-0.156	-0.167
Age	0.082	-0.044	-0.154	0.278	-0.193	-0.003	-0.156	1.000	0.329
Strength	0.498	0.135	-0.106	-0.290	0.366	-0.165	-0.167	0.329	1.000

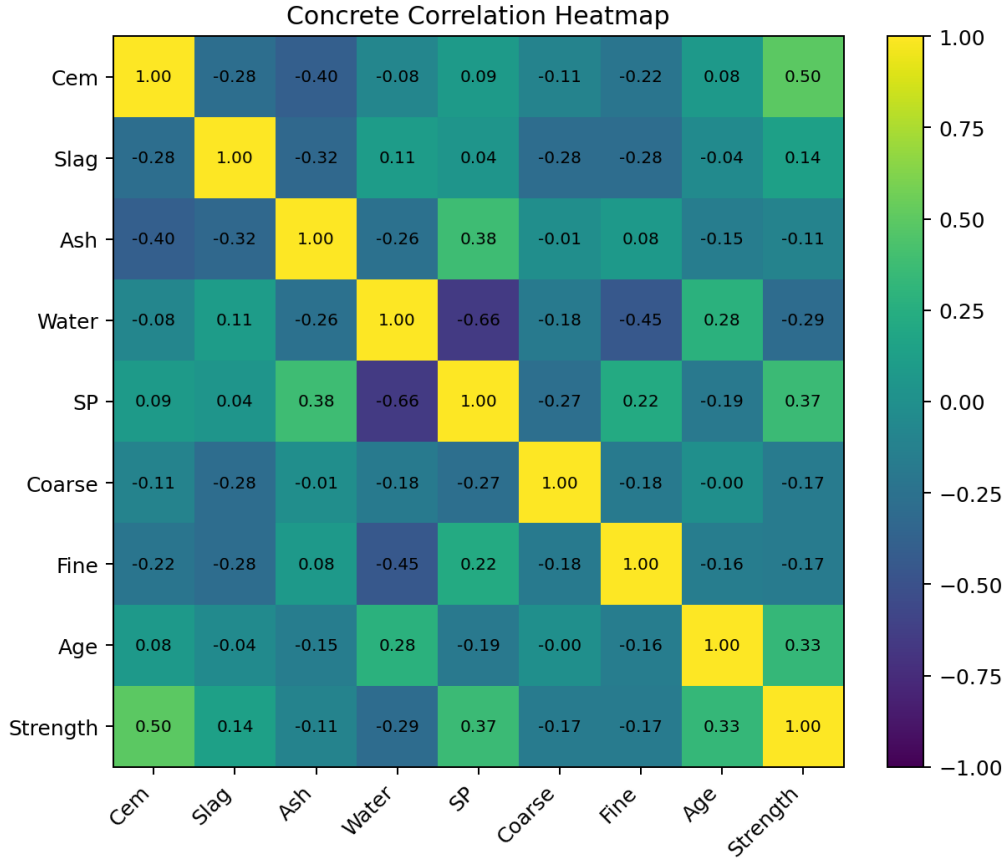


Figure 4: Concrete correlation heatmap. Cement and superplasticizer have the strongest absolute correlations with compressive strength.

3.4 Simple Regression Using Cement

Simple linear regression was performed using both Python’s `statsmodels` package and ScalaTion. Both implementations produced essentially identical coefficient estimates. The fitted model was

$$\widehat{\text{Strength}} = 13.4425 + 0.07958(\text{Cement}),$$

where cement content is measured in kg/m^3 and compressive strength is measured in MPa.

A 1 kg/m^3 increase in cement content is associated with an estimated increase of approximately 0.0796 MPa in compressive strength. The slope was statistically significant ($p < 0.001$). ScalaTion reported $R^2 = 0.247837$ and adjusted $R^2 = 0.247105$, consistent with the `statsmodels` results. Cement alone therefore explained approximately 24.8% of the observed variation in compressive strength. ScalaTion reported an RMSE of approximately 14.48 MPa.

Figure still required: export the observed-versus-fitted plot for Cement and save it as `figures/concrete_cement_fit.png`.

Figure 5: Placeholder for the required Cement observed-versus-fitted plot.

3.5 Simple Regression Using Superplasticizer

The simple regression using superplasticizer produced essentially identical results in ScalaTion and `statsmodels`. The fitted equation was

$$\widehat{\text{Strength}} = 29.4660 + 1.02373(\text{Superplasticizer}),$$

where superplasticizer content is measured in kg/m^3 .

A 1 kg/m^3 increase in superplasticizer content is associated with an estimated increase of approximately 1.024 MPa in compressive strength. The slope was statistically significant ($p < 0.001$). ScalaTion reported $R^2 = 0.134014$ and adjusted $R^2 = 0.133171$, matching the `statsmodels` results to rounding precision. Superplasticizer alone explained approximately 13.4% of the observed variation in compressive strength. ScalaTion reported an RMSE of approximately 15.54 MPa.

Figure still required: export the observed-versus-fitted plot for Superplasticizer and save it as `figures/concrete_superplasticizer_fit.png`.

Figure 6: Placeholder for the required Superplasticizer observed-versus-fitted plot.

3.6 Concrete Discussion

Both selected predictors exhibited positive and statistically significant linear relationships with concrete compressive strength, and ScalaTion and `statsmodels` produced essentially identical results. Cement provided the stronger simple regression model, with $R^2 \approx 0.248$, compared with $R^2 \approx 0.134$ for superplasticizer. This ordering is consistent with the correlation analysis. However, neither predictor individually explained the majority of the variation in strength, indicating that compressive strength depends on multiple mixture components and curing conditions.

4 Airfoil Self-Noise

The Airfoil Self-Noise dataset contains 1,503 observations, five predictor variables, and the response variable scaled sound pressure level in dB.

4.1 Preprocessing

All six original dataset variables were numeric and no missing values were detected. Therefore, no string encoding, imputation, or row deletion was required.

The $1.5 \times IQR$ rule identified 86 potential outliers for frequency, 30 for angle of attack, 124 for displacement thickness, and 4 for sound pressure level. No IQR outliers were identified for chord length or free-stream velocity. These observations were retained because the IQR rule identifies statistically unusual observations but does not show that the observations are incorrect. Removing valid observations could discard information from less common experimental conditions.

The notebook had also retained a generated prediction column named `y_hat` in some saved outputs because cells had been executed out of order. Since `y_hat` is not an original Airfoil variable, it is excluded from the statistical summary and correlation analysis in this report.

4.2 Statistical Summary

Table 5: Airfoil Self-Noise descriptive statistics.

Variable	Count	Mean	SD	Min	Q1	Median	Q3	Max
Frequency	1503	2886.381	3152.573	200.000	800.000	1600.000	4000.000	20000.000
Angle of Attack	1503	6.782	5.918	0.000	2.000	5.400	9.900	22.200
Chord Length	1503	0.137	0.094	0.025	0.051	0.102	0.229	0.305
Free-Stream Velocity	1503	50.861	15.573	31.700	39.600	39.600	71.300	71.300
Displacement Thickness	1503	0.011	0.013	0.000	0.003	0.005	0.016	0.058
Sound Pressure Level	1503	124.836	6.899	103.380	120.191	125.721	129.996	140.987

Sound pressure level has a mean of 124.836 dB and a standard deviation of 6.899 dB. Frequency spans a particularly large range, from 200 Hz to 20,000 Hz.

4.3 Correlation Analysis and Predictor Selection

Frequency has the strongest absolute correlation with sound pressure level ($r = -0.391$), followed by displacement thickness ($r = -0.313$). These two predictors were therefore selected for simple regression.

Table 6: Airfoil Self-Noise Pearson correlation matrix.

	Freq	AoA	Chord	Vel	Thick	SPL
Freq	1.000	-0.273	-0.004	0.134	-0.230	-0.391
AoA	-0.273	1.000	-0.505	0.059	0.753	-0.156
Chord	-0.004	-0.505	1.000	0.004	-0.221	-0.236
Vel	0.134	0.059	0.004	1.000	-0.004	0.125
Thick	-0.230	0.753	-0.221	-0.004	1.000	-0.313
SPL	-0.391	-0.156	-0.236	0.125	-0.313	1.000

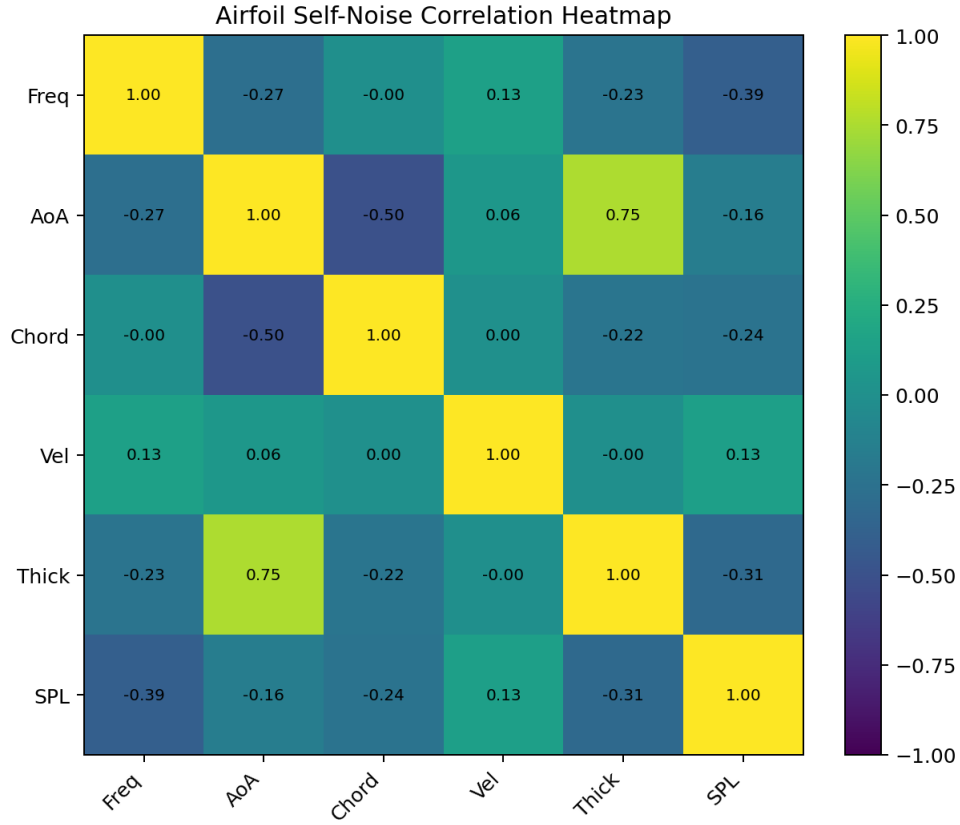


Figure 7: Airfoil Self-Noise correlation heatmap after excluding the generated `y_hat` column.

4.4 Simple Regression Using Frequency

Both software implementations produced essentially identical coefficient estimates. The fitted model was

$$\widehat{\text{SPL}} = 127.3037 - 0.000855(\text{Frequency}).$$

The negative slope indicates an inverse association between frequency and sound pressure level. An increase of 1,000 Hz is associated with an estimated decrease of approximately 0.855 dB in sound pressure level. The slope was statistically significant ($p < 0.001$).

ScalaTion reported $R^2 = 0.152655$ and adjusted $R^2 = 0.152091$, which agreed with the corresponding `statsmodels` results. Thus, frequency alone explained approximately 15.3% of the observed variability in sound pressure level.

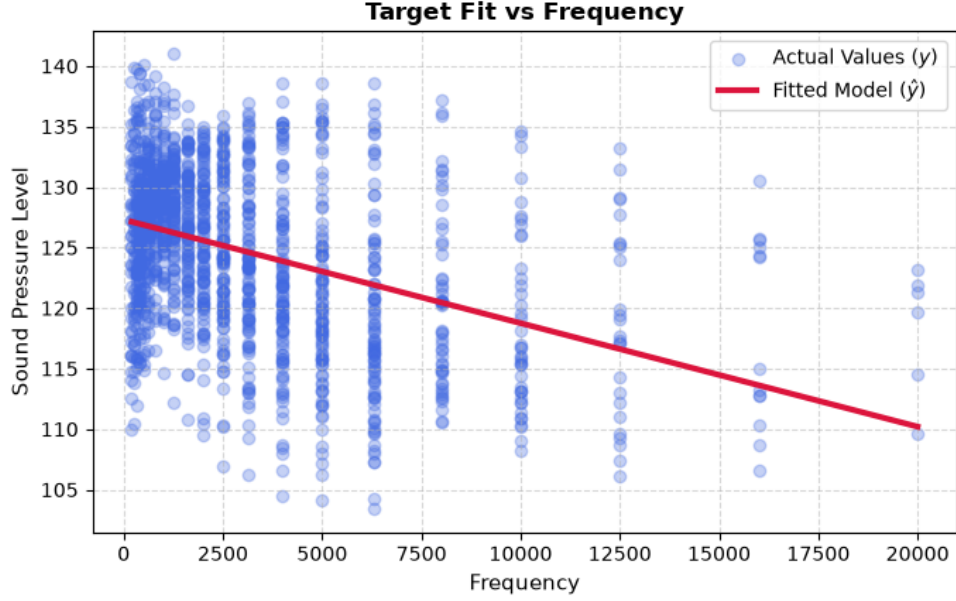


Figure 8: Observed sound pressure levels and fitted values versus frequency.

4.5 Simple Regression Using Displacement Thickness

The fitted equation was

$$\widehat{\text{SPL}} = 126.6632 - 164.0275(\text{Displacement Thickness}),$$

where displacement thickness is measured in meters.

The slope of -164.0275 dB per meter is equivalent to approximately -0.164 dB per millimeter. Therefore, a 1 mm increase in displacement thickness is associated with an estimated decrease of approximately 0.164 dB in sound pressure level. The coefficient was statistically significant ($p < 0.001$).

ScalaTion reported $R^2 = 0.097762$ and adjusted $R^2 = 0.097161$, again matching the `statsmodels` results to rounding precision. Displacement thickness alone therefore explained approximately 9.8% of the observed variability in sound pressure level.

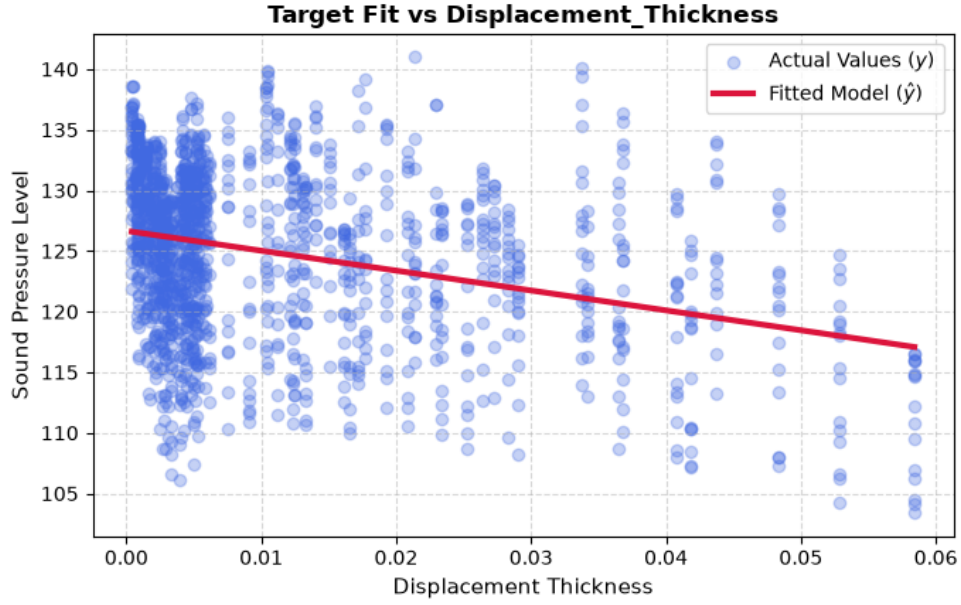


Figure 9: Observed sound pressure levels and fitted values versus displacement thickness.

4.6 Airfoil Discussion

Both selected predictors had statistically significant negative relationships with sound pressure level, and both software packages produced essentially identical results. Frequency provided the stronger simple regression model, with $R^2 \approx 0.153$ compared with approximately 0.098 for displacement thickness. Nevertheless, neither predictor explains a large proportion of the response variability by itself, suggesting that sound pressure level is influenced by additional experimental variables.

5 Overall Conclusion

Across all three datasets, the predictors selected from the correlation analysis were used consistently in both software implementations. `ScalaTion` and `statsmodels` produced essentially identical fitted coefficients and goodness-of-fit values for the six simple regression relationships.

The strongest individual simple regression among the analyses was vehicle weight for AutoMPG, which explained approximately 69.3% of MPG variation. The Concrete and Airfoil models had lower R^2 values, indicating that their responses are less adequately explained by one predictor at a time. The results illustrate both the usefulness and the limitations of simple linear regression: a statistically significant relationship may exist even when a single predictor leaves substantial response variation unexplained.